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In [1]: import numpy as np
import pandas as pd
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In [2]: dict1 = {'Name': ['Priyang', 'Aadhya', 'Krisha', 'Vedant', 'Parshv',
                        'Mittal', 'Archana'],
                'Marks': [98, 89, 99, 87, 90, 83, 99],
                'Gender': ['Male', 'Female', 'Female', 'Male', 'Male',
                          'Female', 'Female']}

df1 = pd.DataFrame(dict1)
df1
```

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Out[2]:
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	Name	Marks	Gender
0	Priyang	98	Male
1	Aadhya	89	Female
2	Krisha	99	Female
3	Vedant	87	Male
4	Parshv	90	Male
5	Mittal	83	Female
6	Archana	99	Female

```
In [4]: # 1. Display Top 3 rows of DataSet
df1.head(3)
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Out[4]:
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	Name	Marks	Gender
0	Priyang	98	Male
1	Aadhya	89	Female
2	Krisha	99	Female

```
In [5]: # 2. Check Last 3 rows of Dataset
df1.tail(3)
```

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Out[5]:
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	Name	Marks	Gender
4	Parshv	90	Male
5	Mittal	83	Female
6	Archana	99	Female

```
In [8]: # 3. Find Shape of our Dataset (no. of rows & columns in dataset)
df1.shape
```

Out[8]: (7, 3)

In [9]: *# 4. Get Information about our dataset Like total number of rows, columns, datatype*
df1.info

Out[9]: <bound method DataFrame.info of

	Name	Marks	Gender
0	Priyang	98	Male
1	Aadhya	89	Female
2	Krishna	99	Female
3	Vedant	87	Male
4	Parshv	90	Male
5	Mittal	83	Female
6	Archana	99	Female

In [14]: *# 5. Check null values in the dataset*
df1.isnull()

Out[14]:

	Name	Marks	Gender
0	False	False	False
1	False	False	False
2	False	False	False
3	False	False	False
4	False	False	False
5	False	False	False
6	False	False	False

In [15]: *# 6. Get overall Statistics about the dataframe*
df1.describe()

Out[15]:

	Marks
count	7.000000
mean	92.142857
std	6.491753
min	83.000000
25%	88.000000
50%	90.000000
75%	98.500000
max	99.000000

In [16]: *# 7. Find unique values from the Gender column*
df1["Gender"].unique()

Out[16]: array(['Male', 'Female'], dtype=object)

In [45]: *# 8. Find the number of unique values from the gender column*
df1["Gender"].nunique()

Out[45]: 2

In [19]: *# 9. Display count of unique values in Gender column*
df1["Gender"].value_counts()

Out[19]: Gender
Female 4
Male 3
Name: count, dtype: int64

In [21]: *# 10. Find total number of students having marks between 90 to 100 (inclusive) using*
df1['Marks'].between(90,100).sum()

Out[21]: np.int64(4)

In [22]: *# 11. Find Average Marks*
df1['Marks'].mean()

Out[22]: np.float64(92.14285714285714)

In [25]: *# 12. Drop the Columns*
df1.drop('Gender',axis=1,inplace=True)

In [26]: *# 13. Print name of the columns*
df1.columns

Out[26]: Index(['Name', 'Marks'], dtype='object')

In [27]: *# 14. Sort the Dataframe as per the marks column*
df1.sort_values(by='Marks',ascending=False)

Out[27]:

	Name	Marks
2	Krishna	99
6	Archana	99
0	Priyang	98
4	Parshv	90
1	Aadhya	89
3	Vedant	87
5	Mittal	83

In [36]: df1.columns = df1.columns.str.strip().str.lower()

```
In [37]: df1.columns
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Out[37]: Index(['name', 'marks'], dtype='object')
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In [43]: ## 15. Display the name & marks of female students  
df1[df1['Gender'] == 'Female'][['Name', 'Marks']]
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Out[43]:
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	Name	Marks
1	Aadhya	89
2	Krishna	99
5	Mittal	83
6	Archana	99

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